Lecture 20: Alternating Series, Absolute Convergence and Conditional Convergence (10.4)

$\boxed{\textbf{Alternating Series}}$ is a series of the form

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots =$$

$$\begin{cases} \displaystyle\sum_{n=1}^{\infty} \underbrace{(-1)^{n+1}}_{\text{sign}} \underbrace{b_n}_{+} = b_1 - b_2 + \cdots + (-1)^{n+1} b_n + \cdots \\ \text{or} \\ \displaystyle\sum_{n=1}^{\infty} \underbrace{(-1)^{n}}_{\text{sign}} \underbrace{b_n}_{+} = -b_1 + b_2 + \cdots + (-1)^{n} b_n + \cdots \end{cases}$$

where $|a_n| = b_n, b_n \geq 0$.

The **signs** of an alternating series changes **every other term**.

$$\sum_{n=1}^{\infty} \underbrace{(-1)^{n+1}}_{\text{sign}} \underbrace{\cos n}_{b_n} \Rightarrow \text{not an alternating series}$$

$\downarrow$ not always positive

positive series

$\sum \frac{1}{n}$ $(p=1)$ div.

Alternating Series Test

If the alternating series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + \ldots, \quad b_n > 0$$

satisfies

and
$$\begin{cases} \text{(i)} \ \lim_{n\to\infty} b_n = 0 \\ \text{(ii)} \ b_{n+1} \le b_n, \quad (b_n \downarrow) \end{cases}$$

then the alternating series converges.

NOTE: For an alternating series to converge, **both conditions** must be satisfied.

ex. $$\sum_{n=1}^{\infty} \frac{\overbrace{(-1)^{n+1}}^{\text{sign}}(n+1)}{n}$$

$b_n > 0$

(i) $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n+1}{n} = 1 \neq 0$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (-1)^{n+1} \frac{n+1}{n}$ DNE $\neq 0$

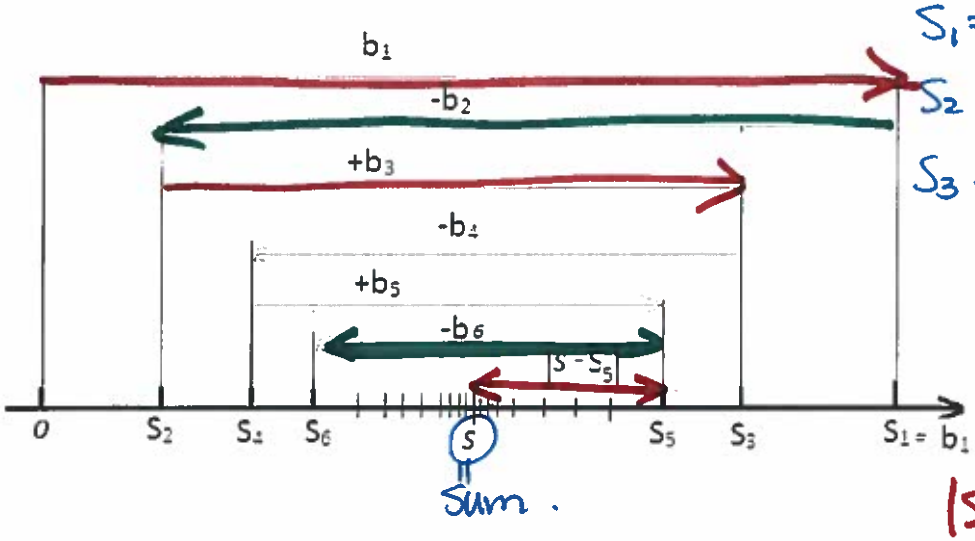
It diverges by TFD.

(Here, the first condition of the alternating series test is not satisfied. In fact, the series diverges, by Test for Divergence).

$$\sum_{1}^{\infty} a_n = \sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + \cdots \quad (b_n > 0)$$

$b_n \downarrow$

$b_n \to 0$



$S_1 = b_1$

$S_2 = b_1 - b_2$

$S_3 = \underbrace{b_1 - b_2} + b_3$

$= S_2 + b_3$

$|S - S_5| \le b_6$

$0 < S < b_1$. $\quad S_{2N} < S < S_{2N+1}$. $\quad |S - S_N| < b_{N+1}$

Determine whether the following series converge, what test do you use?

1. Alternating Harmonic Series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$

$b_n = \frac{1}{n}$

(i) $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n} = 0$

(ii) $b_n \downarrow \qquad b_n = \frac{1}{n} > b_{n+1} = \frac{1}{n+1}$

$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ conv. by A.S.T.

Estimaing Sums

Let s be the sum of a convergent alternating series that satisfies both conditions of the test, and let S_N be the Nth partial sum. Then the **error** of estimation for a partial sum S_N is

sum

$$\boxed{|error| = |R_N| = |s - S_N| \leq \boxed{b_{N+1}}.}$$

In other words, the **error** committed when we approximate s by S_N is no bigger than the size of the first omitted term b_{N+1}.

(This rule doesn't apply to any other types of series.)

ex. Given a convergent alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^4}$,

1. Find the approximation of the sum by adding the first 4 terms. How close is this approximation to the true value?
2. How many terms do I need to add in order to estimate the series within 0.001?

1. Sum $\sim$ $S_4 = \frac{1}{1^4} - \frac{1}{2^4} + \frac{1}{3^4} - \frac{1}{4^4} \sim 0.9459$

$|S - S_4| \leq b_5 = \frac{1}{5^4} \sim 0.0016$

2. $|S - S_N| \leq b_{N+1} = \frac{1}{(N+1)^4} \leq 0.001$

$n=5$: $|S - S_5| \leq \frac{1}{6^4} \sim 0.0007$ ✓

need 5 terms.

Def. A series $\sum_{n=1}^{\infty} a_n$ is **absolutely convergent** if the series $\sum_{n=1}^{\infty} |a_n|$ converges.

Thm. Absolute Convergence Implies Convergence

If $\sum |a_n|$ converges, then $\sum a_n$ also converges.

Def. If $\sum a_n$ converges, but $\sum |a_n|$ diverges, then it converges **conditionally**.

ex. $\sum \frac{1}{n}$ $(p=1)$ div. $\sum \frac{1}{n^2}$ $(p=2)$ conv.

ex. $\sum \frac{(-1)^n}{n}$ conv. by AST. $\sum \frac{(-1)^n}{n^2}$ conv. by AST.

$\Downarrow$ $\Downarrow$

$\sum \left|\frac{(-1)^n}{n}\right| = \sum \frac{1}{n}$ div. $\sum \left|\frac{(-1)^n}{n^2}\right| = \sum \frac{1}{n^2}$ conv.

$\Downarrow$ $\Downarrow$

$\sum \frac{(-1)^n}{n}$ conv. cond $\sum \frac{(-1)^n}{n^2}$ conv. abs.

NYTI:

1. For what values of p is the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ convergent?

2. Given a convergent alternating series how many terms do I need to add so that $|\textbf{error}| \leq 0.1$? .

(a) $\sum_{n=1}^{\infty} (-1)^{n-1} n e^{-n}$

(b) $\sum_{n=1}^{\infty} \frac{n}{(-2)^{n-1}}$ (Conv)

$$= \sum_{n=1}^{\infty} \frac{n}{[(-1)\cdot 2]^{n-1}}$$

3. Does $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ converge absolutely? Explain.

$$= \sum_{n=1}^{\infty} \frac{n}{(-1)^{n-1} \cdot 2^{n-1}}$$

$$= \sum_{n=1}^{\infty} (-1)^{n-1} \cdot \frac{n}{2^{n-1}}$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n^n} = \sum_{n=1}^{\infty} \frac{1}{n^n}$$

Determine the convergence:

3. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n n^n}{n!}$ (Div.)

$\cos(\pi) = -1$ $\quad \cos(2\pi) = 1$ $\quad \cos(3\pi) = -1$ $\quad \cos(4\pi) = 1$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{\sin(n\pi/2)}{n!}$ (conv.)

$$= \frac{1}{1!} - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} + \cdots$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)!}$$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^n}$ (conv.)

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{n^n}$$

6. $\displaystyle\sum_{n=1}^{\infty} (-1)^{n-1} \frac{e^{1/n}}{n}$ (conv.)

7. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$ (TFD)

8. $\displaystyle\sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2}\right)$ (TFD)

div. by TFD.

$$= \sin\left(\frac{\pi}{2}\right) + \underbrace{\sin\left(\frac{2\pi}{2}\right)}_{n=2} + \sin\left(\frac{3\pi}{2}\right) + \underbrace{\sin\left(\frac{4\pi}{2}\right)}_{n=4} + \cdots$$

$$= 1 + 0 - 1 + 0 + - -$$

$$= \sum_{n=1}^{\infty} (-1)^{n+1} \cdot 1 \quad \boxed{b_n = 1}$$

NYTI:

1. For what values of p is the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ convergent? $(p>0)$ $\quad b_n = \frac{1}{n^p}$

(i) $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n^p} = 0$ if $p>0$

(ii) b_n is dec. if $p>0$.

Therefore, if $p>0$, $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ converges by AST.

2. Given a convergent alternating series how many terms do I need to add so that $|\textbf{error}| \leq 0.1$?

$= \sum_{1}^{\infty} \frac{n}{(-1)^{n-1} 2^{n-1}} = \sum_{1}^{\infty} (-1)^{n-1} \cdot \frac{n}{2^{n-1}}$

(a) $\sum_{n=1}^{\infty} (-1)^{n-1} n e^{-n}$

$|\text{error}| \leq b_{N+1} = \frac{N+1}{e^N} \leq 0.1$

$N=3$.

(b) $\sum_{n=1}^{\infty} \frac{n}{(-2)^{n-1}}$ (Conv)

$|\text{error}| \leq b_{N+1} = \frac{N+1}{2^N} \leq 0.1$

$N=7$

3. Does $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ converge absolutely? Explain.

Yes!

$\left|\frac{\cos n}{n^2}\right| \leq \frac{1}{n^2}$ and $\sum \frac{1}{n^2}$ conv. $(p=2)$ $\Rightarrow \sum_{1}^{\infty} \left|\frac{\cos n}{n^2}\right|$ conv. by the comparison test.

$\Rightarrow \sum_{1}^{\infty} \frac{\cos n}{n^2}$ conv. abs.

Determine the convergence:

Div by TFD.

3. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n n^n}{n!}$ (Div.) $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n^n}{n!} = \infty$

So $\lim_{n\to\infty} a_n = \lim_{n\to\infty} (-1)^n b_n$ DNE $\neq 0$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{\sin(n\pi/2)}{n!}$ (conv.) conv. by AST

$= \displaystyle\sum_{1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)!}$ (see #8)

$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{(2n-1)!} = 0$ and $b_n \downarrow$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^n}$ (conv.) conv. by AST

$= \displaystyle\sum_{1}^{\infty} \frac{(-1)^n}{n^n}$. $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n^n} = 0$ and $b_n \downarrow$

6. $\displaystyle\sum_{n=1}^{\infty} (-1)^{n-1} \frac{e^{1/n}}{n}$ (conv.) conv. by AST.

Let $f(x) = \dfrac{e^{1/x}}{x}$

$\lim_{x\to\infty} \dfrac{e^{1/x}}{x} \overset{L'H}{=} 0$

and $f'(x) = \dfrac{e^{1/x}(e^{-1}-1)}{x^2} < 0$

7. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$ (TFD) div. by TFD.

$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{3n}{4n-1} = \frac{3}{4}$

So $\lim_{n\to\infty} a_n = \lim_{n\to\infty} (-1)^n b_n$ DNE $\neq 0$

8. $\displaystyle\sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2}\right)$ (TFD) div. by TFD.

$= \underset{(n=1)}{1} + \underset{(n=2)}{0} + \underset{(n=3)}{(-1)} + \underset{(n=4)}{0} + 1 + 0 + (-1) + 0 + \cdots$

$= \displaystyle\sum_{1}^{\infty} (-1)^{n-1}$ $\lim_{n\to\infty} a_n = \lim_{n\to\infty} (-1)^{n+1}$ DNE $\neq 0$